

ADITYA NAIDU

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EDUCATION

ACROPOLIS INSTITUTE OF TECHNOLOGY AND RESEARCH (AITR) Indore, MP

B.Tech in Computer Science & Engineering (AI/ML), CGPA: 6.1

A.P. NARMADA HR. SEC. SCHOOL (MP Board) Jabalpur, MP

Higher Secondary Certificate (10+2), Science — 72.8%,

ST. ALOYSIUS SCHOOL (CBSE) Jabalpur, MP

Secondary School Certificate (10th) — 79.4%

TECHNICAL SKILLS

Languages: Python, SQL, GML Code

Data Science & ML: NumPy, Pandas, scikit-learn, PyTorch, OpenCV, YOLO, ONNX Runtime, TensorRT, LangGraph

Tools & Platforms: Git/GitHub, Jupyter, FastAPI, Flask, PyQt5, ROS2 (Humble), NVIDIA Jetson, Linux, IPG CarMaker

Technologies: Computer vision, sensor data processing, time-series smoothing, PID control, ML, ADAS

PROJECTS

Autonomous Baja: Team Acroracerz (2025-2025)

- **Object Detection:** Built a real-time 15-class detector for signs, lights, pedestrians, and vehicles; benchmarked YOLOv8n/v10n/v11n; deployed a detector + specialist classifier with TensorRT on Jetson Orin NX at 50 FPS.
- **Lane Keeping Assist:** Developed lane segmentation with PID control to keep the vehicle centered.
- **Automatic Emergency Braking:** Built radar-based obstacle detection that triggers automatic braking.
- **Endurance:** Integrated LKA and AEB into a unified Endurance Arbitrator combining lateral and longitudinal control.

Rulebook Assistant: RAG Q&A over SAE Baja Rulebook

- Built a retrieval-augmented Q&A system over the SAE Baja rulebook (N pages) using bge-small embeddings, ChromaDB, and a hybrid BM25 + vector retriever, returning answers with rule-ID and page citations.
- Evaluated on a hand-labelled set of N questions: hybrid retrieval raised recall@5 from X% to Y% over vector-only search; served via FastAPI with median latency of Z s.

Vitesse — Industrial ROS2 Inspector & Performance Monitor

- Developed a PyQt5-based ROS2 monitoring dashboard for real-time topic, publisher, subscriber, and message inspection.
- Implemented Hz, latency, cycle-time, and throughput analysis with dynamic topic discovery to identify communication bottlenecks.
- Deployed the project to PyPI for public distribution.

Palette: Multi-Agent Orchestration Framework (2025)

- Built a Python framework on LangGraph that abstracts multi-agent setup (models, prompts, graph wiring, termination), cutting setup time by 70-80%.
- Implemented a round-robin StateGraph with Primary and Critic agents, conditional-edge termination (external or mention-based), and per-agent switchable LLM providers (OpenAI, Anthropic, Azure).

LEADERSHIP & EXPERIENCE

Team Acroracerz — Captain & AI/ADAS Developer Feb 2026 – Present

aBAJA SAE India, Vehicle #264001

- Led the migration of the team's vehicle software stack from ad hoc scripts to a distributed, ROS2-based architecture.
- Developed core autonomy modules (Lane Keep Assist, Traffic Light & Sign Detection) and an Endurance Arbitrator fusing lateral and longitudinal control commands, deployed to an NVIDIA Jetson Orin NX with sensor fusion (camera, radar, GNSS, IMU) at 50 Hz.
- Represented the team in official communications; secured Drive-by-Wire Event Winner honors in Phase 3 of aBAJA SAE India 2025 (Global Automotive Research Centre, Chennai), contributing to an AIR 1 national ranking

ACHIEVEMENTS

- **aBAJA SAE India 2026:** 2nd Runner-Up (3rd Overall) as Team Captain; Winner — AEB Event; 2nd Place — Software Design; 2nd Runner-Up — Lane Keep Assist
- **aBAJA SAE India 2025:** Winner — Drive-by-Wire Design Event; contributed to AIR 1 national ranking